

Digitization and social inequality

Objective of the lecture: To know the details on how internet use, digital technologies, platforms, ICTs transforms social inequality by ethnicity, gender, and age

1. What digitization does to inequality?
2. How digitization transforms inequality - Key Mechanisms
 - 2.1 Digital Inequality by Ethnicity
 - 2.2 Digital Inequality by Gender
 - 2.3 Digital Inequality by Age
3. Digital inclusion challenges in Bangladesh

1. What digitization does to inequality?

Digitization has become a central feature of modern life, with billions of people using digital platforms, apps, and ICT services. Digitization is not socially neutral; rather, it transforms existing social inequalities and can also generate new digital inequalities. Verwiebe et al. conducted a PRISMA-based systematic review of 281 articles (2000–2020) focusing on inequality dimensions of ethnicity, gender, and age.

No definitive outcome (increase or decrease in inequality) can be universally stated because empirical results vary by context, dataset, and mechanism. However, a general tendency exists: more studies find inequality is reproduced or reduced rather than increased. Distribution of inequality among article published shows that –

- 46% of studies reach unclear conclusions

- 37% show reproduction of inequality

- 12% show reduction of inequality

- Only 4% show an increase in inequality

2. How digitization transforms inequality - Key Mechanisms

Inequality outcomes are not fixed; rather, digitization can lead to reproduction, reduction, or sometimes increase, depending on context.

1) Digital Divide: Unequal Access and Unequal Use: Digital inequality emerges from differences in access to and use of ICT and digital technologies. Even though the gap has diminished over time, inequalities remain visible and continue to affect social participation.

2) Digital Divide Has “Levels” (From Access to Outcomes): Digital inequality is not limited to internet access only. It also involves differences in digital skills, usage practices, and outcomes, which create unequal benefits from digitization across social groups.

3) Feedback Effects: Digital Inequality Reinforces Existing Inequality: A major mechanism is the feedback loop: unequal access and use of digital technologies are connected to pre-existing inequalities in education and income. This produces a cycle where digital inequalities reinforce already unequally distributed social resources.

2. How digitization transforms inequality - Key Mechanisms

4) *Unequal Social Participation Through Digitization*: Unequal access and use leads to unequal levels of social participation. Digitization therefore affects inequality not only economically, but also socially—through who participates in networks, civic engagement, and communication opportunities.

5) *Discrimination and Exclusion Inside Digital Spaces*: Digital spaces are not neutral. In many studies, patterns of exclusion and discrimination operate inside online platforms, especially in relation to gender and ethnicity.

6) *Activism, Counter-Narratives, and Social Policy Influence*: Digitization can also reduce inequality by enabling the formation of political counter-narratives, online activism, and new possibilities of influencing social policy. These mechanisms can make discrimination more visible and challenge existing inequality structures.

7) *Integration of Migrants and Refugees Through Digital Tools*: ICT provides new opportunities for migrants and refugees by strengthening integration into social networks and labor markets. This mechanism is especially relevant to ethnic inequality reduction in some contexts.

2. How digitization transforms inequality - Key Mechanisms

8) *Social Inclusion of Older Adults: Reducing Isolation:* Digitization may reduce inequality by supporting older adults through increased social integration, strengthening social networks, and increasing life satisfaction and individual well-being.

9) *Unequal Benefits Even Within Groups:* Even when digitization improves well-being (e.g., among older adults), benefits are not equally distributed. Positive effects depend on other inequality factors such as education, income, and gender, which means digitization can reduce isolation but still reproduce inequality internally.

10) *Overall Pattern of Inequality Outcomes (Reproduction vs Reduction vs Increase):* Digitization-related inequality dynamics are mixed, with many studies showing unclear effects (46%). Among clear outcomes, the largest share finds reproduction (37%), followed by reduction (12%), while only around 4% report increased inequality.

2.1 Digital Inequality by Ethnicity

- 1) *Ethnicity remains a strong dividing line:* Ethnicity continues to be a key marker of digital inequality. Ethnic differences differentiate access to ICT and usage opportunities due to unequal material, cultural, and social resources.
- 2) *Digitization reproduces ethnic inequality through discrimination:* Ethnic minorities face disadvantages in digital economic activities and online representation. For instance, research using Airbnb data shows ethnic minority hosts receive lower prices, indicating structural discrimination within platform markets.
- 3) *Digitization can reduce ethnic inequality through mobilization and integration:* Digital activism and counter-narratives can reduce exclusion. Social movements and digital spaces can amplify political participation among ethnic minorities. In addition, digital technologies support migrant and refugee integration into social networks and labor markets.

2.2 Digital Inequality by Gender

- 1) *Gender digital divide depends on location:* A gender gap in ICT access remains especially in the Global South, where women have lower access and lower ICT agency. However, in many Western societies and younger age groups, the gender-specific divide has become much smaller or nearly absent compared to earlier decades.
- 2) *Gender inequality is reproduced through online harassment and exclusion:* Even when access improves, digital spaces reproduce gender inequality through sexism, harassment, and misogynistic discrimination. Studies using social media data show widespread misogynistic content and structural exclusion of women in gaming and digital labor markets.
- 3) *Digital labor market restructuring may harm women:* Digitization may disproportionately reduce female jobs because women are more likely to occupy routine jobs that can be automated or replaced by digital technologies.

2.3 Digital Inequality by Age

- 1) *Age-based digital divide tends to decrease:* Digital divide by age tends to decline, meaning older populations increasingly engage with ICT.
- 2) *ICT improves social inclusion and reduces loneliness:* A major set of studies shows ICT can reduce loneliness and social isolation among older adults. Digital neighborhood platforms and social networks create communication infrastructure that supports integration and social capital.
- 3) *Benefits are unequally distributed among older people:* Even when ICT improves well-being, benefits are determined by education, income, and gender. ICT advantages are therefore not evenly distributed within the elderly population.

3. Digital inclusion challenges in Bangladesh

- 1) *Techno-centric and narrow policy framing (too focused on access)*: The National ICT Policy is described as ambiguous and techno-centric, relying heavily on digitization and infrastructure expansion rather than addressing deeper inclusion barriers. The policy focuses mainly on physical access and connectivity, but does not sufficiently integrate social realities such as poverty, inequality, and exclusion.
- 2) *Weak contextualization of Bangladesh's socio-economic realities*: Bangladesh has high inequality (income, gender, rural–urban), large rural population (66.2%), structural barriers like poor infrastructure, unemployment, political instability and marginalization, climate vulnerability and displacement risks (floods, cyclones, disasters)
- 3) *Rural–Urban digital divide remains severe*: Even with high telecom coverage, internet access is ~8 times lower in rural areas than in urban areas, rural communities face unstable electricity, remoteness, and low awareness of ICT benefits

3. Digital inclusion challenges in Bangladesh

- 4) *Policy vagueness and unclear concept definitions:* Lacks clear definitions of key concepts like ICT, digital divide, e-governance, and inclusion, uses terms like “digital divide” without contextual explanation, presents goals, strategies, and action items in a scattered, unclear way
- 5) *Insufficient focus on disadvantaged groups:* Although National ICT Policy (NIP) claims it will include – low-income groups, ethnic minorities, persons with disabilities and special needs, the policy does not provide clear targeting or practical pathways for reaching the poorest and most excluded groups.
- 6) *Inconsistent and overlapping strategies:* The policy repeatedly uses infrastructure development as the main “solution,” even when discussing skills or participation. This creates overlapping means, unclear goal-means relationships, inconsistent outcomes.
- 7) *Limited citizen-centred ICT use and engagement:* ICT engagement in the NIP is mostly framed around government authority, administrative efficiency, economic growth (agriculture, SMEs, environment, health) But the policy neglects civic participation, cultural inclusion, political engagement, citizens’ everyday ICT practices

3. Digital inclusion challenges in Bangladesh

8) *Skills and digital literacy are underdeveloped*: The NIP has no clear objective specifically focused on ICT skills. It includes ICT curriculum integration, skill assessments, industry–university collaborations, public service exam ICT requirements. But the policy is heavily market-oriented, benefiting skilled/educated groups, and doesn't ensure broad-based skills development for poor and marginal citizens.

9) *Language divide = digital divide*: People proficient in English can access more digital content, and those who only know Bangla face disadvantage, especially in remote areas.

10) *Gender disparity and unequal education quality*: Digital inclusion is further blocked by gender inequality in education, resources, safety, and mobility, wide regional variation in school quality due to centralised governance and weak infrastructure